NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2023-2024

IE2104 – DIGITAL ELECTRONICS

November / December 2023 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 5 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. (a) Draw a CMOS circuit for the logic function $z(a,b,c,d,e) = (e+b)\cdot\bar{d}\cdot(a+c)$ with minimum number of transistors.

(7 Marks)

(b) Given a logic function $y(a,b,c,d) = \sum m(0,1,3,8,10,14) + x(2,4,9,11,12)$, where $x(2,4,9,11,12)$ are don't-care terms.

(i) Obtain the minimized sum-of-products (SOP) expression for $y(a,b,c,d)$.

(ii) Obtain the minimized product-of-sums (POS) expression for $y(a,b,c,d)$.

(10 Marks)

(c) Find the radices x and y such that both of the following two equations are valid.

$$57_x - 1111_2 = 22_y + 15_{16}$$

$$44_y - 13_8 = 11_x + 11_{10}$$

(8 Marks)

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2. (a) Implement the logic function $f(a, b, c) = \sum m(0,3,6,7)$ with the decoder given in Table 1 and ONE logic gate.

Table 1

Input			Output							
I_2	I_1	I_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	0	1
0	1	0	1	1	1	1	1	0	1	1
0	1	1	1	1	1	1	0	1	1	1
1	0	0	1	1	1	0	1	1	1	1
1	0	1	1	1	0	1	1	1	1	1
1	1	0	1	0	1	1	1	1	1	1
1	1	1	0	1	1	1	1	1	1	1

(8 Marks)

(b) Derive the logic function *g(x, y, z)* in the maxterm form $\Pi M(...)$ for the logic circuit given in Figure 1, which consists of a multiplexer (MUX) and a priority encoder. The function tables of the MUX and the encoder are given in Table 2 and Table 3, on page 3, respectively.

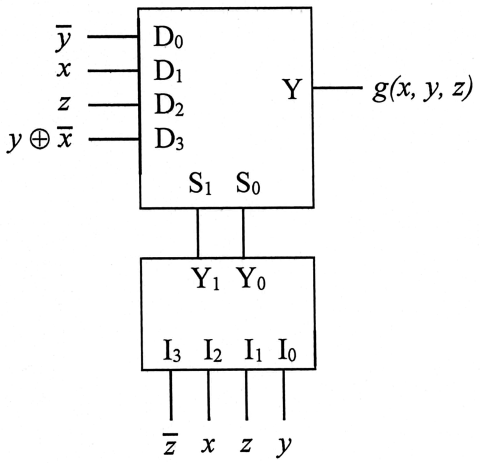


Figure 1

Note: Question No. 2 continues in page 3.

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Table 2

Select		Output
S_1	S_0	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

Table 3

Input				Output	
I_3	I_2	I_1	I_0	Y_1	Y_0
1	X	X	X	1	1
0	1	X	X	1	0
0	0	1	X	0	1
0	0	0	X	0	0

(12 Marks)

(c) Implement the logic function $f(a, b, c) = \sum m(0,3,6,7)$ in Part (a) with ONLY the logic circuit in Figure 1. Use the logic symbol in Figure 2 for the logic circuit. Show your working and label clearly all the inputs and output.

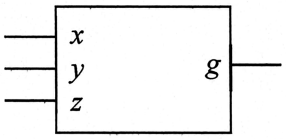


Figure 2

(5 Marks)

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3. (a) Draw a NOR-based SR latch. Tabulate the characteristic table and the excitation table for this latch. Show the circuit of a positive level trigger T flip-flop based on the NOR-based SR latch.

(10 Marks)

(b) You are required to design a sequence detector based on Moore model state machine to detect an input sequence of 1,0,1, with overlapping of input '1' allowed, using T flip-flops and necessary logic gates. Given the input data, X, of 1,1,0,1,0,0,1,1,0,1,0,1,1,0, and output Z, assuming the initial state is 00. Execute your design by answering the following questions:

(i) Draw the state diagram.

(ii) Tabulate the state table.

(iii) Plot the K-maps and write the flip-flop input equations.

(iv) Realize the circuit using T flip-flops and NAND gates.

(v) Show the output sequence of this detector for the given input sequence X.

(12 Marks)

(c) Briefly discuss how to avoid setup time and hold time violations.

(3 Marks)

4. (a) Given a 64M × 1 DRAM. Each memory cell needs to be refreshed every 50ms and the refresh time is 40ns. Determine

(i) the number of rows and columns in the memory array configuration,

(ii) the refresh period for each row,

(iii) the number of refresh cycles which can be accommodated within one refresh period, and

(iv) briefly discuss the reason of arranging the memory array in a square.

(8 Marks)

Note: Question No. 4 continues in page 5.

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(b) A sequence generation circuit based on D flip-flops is given in Figure 3, where Z is the sequence output. Analyze the circuit based on the following steps:

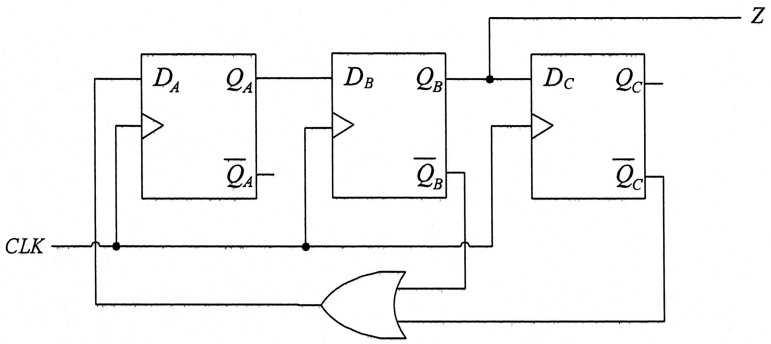


Figure 3

(i) Write the input functions of the flip-flops.

(ii) Plot the K-maps of the input functions.

(iii) Tabulate the state tables.

(iv) Draw the state diagram.

(v) Determine the sequence at the output Z assuming the initial state at the outputs of the flip-flops, $Q_AQ_BQ_C$ are 000.

(13 Marks)

(c) Briefly discuss the difference between Noise Margin (NM) and Signal-to-Noise ratio (SNR).

(4 Marks)

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.

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